Worksheet 4B: What Derivatives and Integrals Tell Us — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. (a) Inc: $x > 0$, Dec: $x < 0$ (b) $x = 0$ (c) 6
2. (a) $3x^2 - 3$ (b) $x = \pm 1$ (c) max at -1, min at 1
3. CPs: $0, \pm 2$; max at 0, min at ± 2
4. $y = 4x - 4$
5. $y = 2x - 2$
6. $(\pm 2, \mp 16)$
7. (a) $20 - 10t$ (b) $t = 2$, max height (c) -10, constant gravity
8. (a) $v = 3t^2 - 12t + 9$, $a = 6t - 12$ (b) $t = 1, 3$ (c) $t = 2$
9. Up: $x > 0$, Down: $x < 0$, Inflection: $x = 0$
10. Inflection at $x = \pm 1$
11. $x = 40$, max profit \$400
12. 25×25 m
13. $t = 3$, height 55 m
14. $p = 125$, revenue \$31,250
15. $C' = 3q^2 - 24q + 60$, min at $q = 4$
16. potential max/min; whether it's max or min

Part B

17. 12
18. 8
19. $\frac{8}{3}$
20. 10

21. 0
22. 0
23. $\frac{32}{3}$
24. 0 (positive and negative areas cancel)
25. 8 m
26. (a) Forward: $t > 2$, Backward: $t < 2$ (b) 0 (c) 4 m
27. 800 L
28. 1125 people
29. (a) $\frac{x^3}{3}$ (b) $\frac{1}{3}, \frac{8}{3}, 9$ (c) x^2
30. $x^3 + 1$
31. $\sqrt{x}$

Part C

32. x^4
33. $3x^2 - x$
34. $\frac{1}{x^2}$
35. 9
36. 18
37. 12
38. 1
39. $\frac{x^4}{4} + C$; x^3; yes
40. $12x - 4$; $6x^2 - 4x + C$; yes
41. Velocity, Acceleration; Acceleration, Position
42. $v(t) = 3t^2 + 2$; $s(t) = t^3 + 2t + 1$

Part D

43. $\frac{1}{6}$
44. $\frac{32}{3}$
45. $\frac{4}{3}$
46. $\frac{1}{2}$
47. $q^* = 10$, surplus $= 100$
48. Gini $= \frac{1/6}{1/2} = \frac{1}{3}$

Part E

49. Min at $w = 3$, $L(3) = 1$
50. $w_1 = 0.4$, $w_2 = 0.72$, $w_3 = 0.976$, approaching 2
51. $\frac{16}{3}$
52. $\frac{4}{3}$
53. $w = \frac{10}{3} \approx 3.33$

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